When to Rerank: Calibrated Safety-Aware Routing for Efficient Retrieval Cascades

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Abstract

Modern retrieval-augmented generation systems often apply expensive hybrid retrieval and Cross-Encoder re-ranking uniformly, despite substantial query-level variation in when re-ranking helps, harms, or is unnecessary. We introduce **B-P-SAFE-AMSR** (*Budgeted Probabilistic Safety-Aware Adaptive Multi-Source Retrieval*), a calibrated safety-aware retrieval router that treats re-ranking as a per-query decision under quality, latency, and harm constraints. For each query, the router estimates the probability of retrieval gain, probability of degradation, expected nDCG improvement, and expected latency, then selects Dense retrieval or Deep Hybrid retrieval using a risk-adjusted utility function. Three operating modes (*lite*, *balanced*, *high_recall*) expose an explicit quality–latency–safety trade-off surface.

Across four BEIR benchmark domains (SciFact, FiQA, NFCorpus, ArguAna), B-P-SAFE-AMSR exhibits dataset-adaptive behaviour: it selectively escalates on SciFact, FiQA, and NFCorpus, retaining substantial fractions of Deep Hybrid quality gain while reducing re-ranking latency, and behaves as a protection/no-benefit controller on ArguAna where always-on re-ranking is not reliably beneficial. In selective-escalation settings, the system saves 31.4%–42.3% latency while retaining up to 88.1% of the quality gain afforded by full re-ranking. In protection/no-benefit settings such as ArguAna, it achieves 55%–61% latency reduction without statistically significant degradation. Performance claims are evaluated using paired t-tests, Wilcoxon signed-rank tests, bootstrap confidence intervals, and permutation tests. These results suggest that calibrated safety-aware routing can improve the quality–cost–safety trade-off frontier of retrieval cascades.

Index Terms

Adaptive retrieval, cascade inference, calibrated classifiers, Retrieval-Augmented Generation, BEIR benchmark, dense retrieval, cross-encoder re-ranking, expected utility, latency optimization.

I. Introduction

Dense retrieval systems built on bi-encoder architectures have become the de facto backbone of modern RAG infrastructure [1], [2]. A FAISS-backed approximate nearest-neighbour index retrieves the top-k candidates from millions of documents in sub-millisecond time, providing a strong quality ceiling for straightforward queries. However, when semantic precision is critical, as in biomedical information retrieval or financial question answering, practitioners augment the dense stage with a Cross-Encoder re-ranker that re-scores all candidates pairwise with a large transformer. This operation is semantically powerful but catastrophically slow: our measurements across four BEIR corpora reveal a mean Cross-Encoder latency of 736.69 ms per query (SciFact, `high_recall` mode), compared to 0.37 ms for the dense stage alone, a latency ratio exceeding $2{,}000\times$.

A. The Static Pipeline Problem

Existing production retrieval pipelines apply this expensive re-ranking *unconditionally*. A query such as "*what is aspirin?*" receives identical computational treatment to "*molecular mechanisms of EGFR inhibitor resistance in non-small-cell lung carcinoma.*" This is deeply suboptimal: easy queries that the dense model already ranks correctly are simply wasting compute, while genuinely hard queries may not be escalated with sufficient depth. The **Pareto Frontier** separating high-latency Cross-Encoder re-ranking from low-latency dense embeddings represents the unsolved design space that this work targets.

B. Cascade Retrieval and Dynamic Routing

The multi-stage cascade retrieval paradigm, formalized by Wang et al. [3] and further analyzed in the context of early-exit strategies by Busolin et al. [4], provides the theoretical scaffold for our approach. In a cascade, cheaper stages serve as *gatekeepers* for expensive stages. The challenge is learning an accurate, *calibrated* gating policy that generalizes across heterogeneous query distributions and out-of-domain corpora.

Our contribution is a *probabilistic, utility-maximizing router* trained on per-query feature vectors derived from dense, BM25, and graph-neighbourhood signals. Unlike classical Learning-to-Rank (LTR) approaches that optimize a global ranking objective, our router operates at the *inference-pipeline level*, deciding *whether* to invoke heavy computation, not how to re-order documents within it. This distinction positions the framework as a lightweight middleware layer that can be dropped in front of any existing retrieval stack.

C. Contributions

This paper makes the following contributions:

- **Probabilistic safety-aware routing formulation.** We frame retrieval-cascade escalation as per-query action selection with explicit gain, harm, latency, and recovery utility components (Equation 1).
- **Calibrated router architecture.** Separate calibrated classifiers for $P(\text{gain})$ and $P(\text{harm})$, a Ridge regression $\hat{\delta}$-nDCG predictor, and a latency estimator, with Dense fallback when expected utility is non-positive.
- **Multi-mode operating frontier.** Three operating profiles (*lite*, *balanced*, *high_recall*) control the quality–latency–safety trade-off via mode-specific λ hyperparameters.
- **Multi-dataset empirical evaluation.** BEIR-style evaluation across four heterogeneous domains showing selective escalation on SciFact, FiQA, and NFCorpus ($p < 0.05$ vs. Dense) and protection/no-benefit behaviour on ArguAna, supported by paired t-tests, Wilcoxon signed-rank tests, bootstrap CIs, and permutation tests.

II. FORMAL ARCHITECTURAL METHODOLOGY

A. Expected Utility Objective

Let $x \in \mathbb{R}^{25}$ be the query feature vector (defined in Section II-D) and let a denote an escalation action drawn from the action space $\mathcal{A} = \{A_0^{\text{Dense}}, A_6^{\text{DeepHybrid}}\}$. The router evaluates the following **Expected Utility** objective for each candidate action:

$$\begin{aligned}\mathcal{U}(a \mid x) = \hat{\delta}(a, x) &- \lambda_{\text{harm}} \cdot P(\text{harm} \mid x, a) \\ &- \lambda_{\text{latency}} \cdot \Delta t(a, x) \\ &- \lambda_{\text{cand}} \cdot n_{\text{cand}}(a) \\ &+ \lambda_{\text{recovery}} \cdot P(\text{gain} \mid x, a)\end{aligned} \tag{1}$$

where:

- $\hat{\delta}(a, x)$ is the predicted nDCG@10 delta from a Ridge regression model trained on labelled training-split queries;
- $P(\text{gain} \mid x, a)$ is the calibrated probability that action a yields a positive nDCG@10 improvement (>0.01) over the dense baseline;
- $P(\text{harm} \mid x, a)$ is the calibrated probability that action a *degrades* retrieval quality ($\Delta\text{nDCG} < -0.01$) relative to the dense baseline;
- $\Delta t(a, x)$ is the predicted per-query latency for action a (ms), from a second Ridge regression model;
- $n_{\text{cand}}(a)$ is the candidate set size for action a (50 for dense, 400 for deep hybrid);
- $\lambda_{\text{harm}}, \lambda_{\text{latency}}, \lambda_{\text{recovery}}, \lambda_{\text{cand}}$ are mode-specific trade-off hyperparameters.

This formulation can be equivalently stated at the aggregate level by substituting calibrated model expectations:

$$\begin{aligned}\mathcal{U}(a \mid x) = P(\text{gain} \mid x) &- \lambda_{\text{harm}} \cdot P(\text{harm} \mid x) \\ &- \lambda_{\text{latency}} \cdot \Delta t(a) \\ &+ \lambda_{\text{recovery}} \cdot \mathcal{R}(a)\end{aligned} \tag{2}$$

where $\mathcal{R}(a) \equiv P(\text{gain} \mid x, a)$ serves as a recovery signal promoting aggressive escalation when the model is confident the query is hard.

The router defaults to the dense baseline A_0^{Dense} (utility anchored at 0.0) and escalates *only* when $\mathcal{U}(A_6^{\text{DeepHybrid}} \mid x) > 0$.

B. Mode-Specific Hyperparameter Profiles

Three operating modes encode distinct compute-quality trade-offs via λ profiles. Table I reports the default hyperparameter values for each mode. The *balanced* and *high_recall* modes additionally undergo grid-search tuning on the validation split before test evaluation.

TABLE I
MODE-SPECIFIC HYPERPARAMETER PROFILES

Mode	λ_{lat}	λ_{harm}	λ_{rec}	τ_{gain}	τ_{harm}	LCB
lite	2×10^{-4}	0.50	0.10	0.60	0.20	on
balanced	5×10^{-5}	0.25	0.25	0.40	0.40	off
high_recall	1×10^{-5}	0.15	0.35	0.25	0.55	off

The *lite* mode applies the strongest harm penalty ($\lambda_{\text{harm}} = 0.50$) and the highest gain threshold ($\tau_{\text{gain}} = 0.60$), making escalation extremely conservative. The *high_recall* mode relaxes all constraints: the lowest latency penalty ($\lambda_{\text{lat}} = 10^{-5}$), a generous harm threshold ($\tau_{\text{harm}} = 0.55$), and a low gain admission gate ($\tau_{\text{gain}} = 0.25$) to maximise quality on hard queries at the cost of higher compute utilization.

C. Calibration Layer

A central architectural commitment is the use of `CalibratedClassifierCV` (scikit-learn) with sigmoid Platt scaling as the classifier for both the *gain* and *harm* probability models. This choice is non-trivial and has a substantial impact on hyperparameter stability.

Definition 1 (Calibration Gap)**.** *A binary classifier is* well-calibrated *if for all* $p \in [0, 1]$*:*

$$P(y = 1 \mid \hat{p}(x) = p) = p, \tag{3}$$

where $\hat{p}(x)$ is the model's predicted probability and y is the ground-truth label.

Uncalibrated logistic regression outputs that emerge from imbalanced training sets (as is common in retrieval scenarios where easy queries dominate) systematically under- or over-estimate true event frequencies. When such raw scores are used directly as $P(\text{gain})$ and $P(\text{harm})$ in Equation (2), the λ surface becomes ill-conditioned: small changes to λ_{harm} produce large, erratic changes to routing decisions. Sigmoid calibration maps raw log-odds to an isotonic frequency-matching curve, providing a stable posterior probability surface across heterogeneous, out-of-domain text fields.

The calibration cross-validation fold count is set as $\text{cv} = \min(3, n_{\text{minority\_class}})$, ensuring graceful degradation to a `PriorProbabilityModel` fallback when the minority class count is insufficient for proper cross-validation (e.g., ArguAna's near-zero-benefit regime).

D. Feature Extraction Pipeline

Each query is represented by a 25-dimensional feature vector $x \in \mathbb{R}^{25}$, extracted during the feature extraction phase of the inference pipeline. The features span four signal families:

a) Lexical Query Features (6 dimensions): Query token count, presence of numeric characters, presence of uppercase abbreviations (regex: `[A-Z]{2,}`), mean per-token IDF score, maximum per-token IDF score, and a lexical specificity score (fraction of tokens exceeding the corpus median IDF). The IDF dictionary is built from a TF-IDF vectorizer fitted to the full corpus with a vocabulary cap of 50,000 terms.

b) Dense Retrieval Features (6 dimensions): Maximum, mean, and standard deviation of the top-50 dense similarity scores; normalised Shannon entropy of the score distribution; and score gaps between rank-1 and rank-5 ($\delta_{1,5}$) and rank-1 and rank-10 ($\delta_{1,10}$). A high $\delta_{1,5}$ gap indicates a confident, separable top result, suppressing escalation.

c) BM25 Lexical Features (5 dimensions): Maximum, mean, standard deviation, normalised entropy, and score gap for the BM25 top-100 candidate scores (normalised by the corpus-level BM25 maximum).

d) Cross-Retriever Disagreement Features (5 dimensions): Jaccard overlap between top-10 dense and BM25 candidates; Jaccard overlap at top-50; Kendall-τ rank correlation over shared candidates; candidate novelty ($1 - J_{50}$); and a source disagreement score combining low Jaccard and low rank correlation. High disagreement is a strong signal that the query is ambiguous and benefits from re-ranking.

e) Graph Neighbourhood Features (3 dimensions): Maximum and mean degree of the top-10 dense candidates in the FAISS-built k-nearest-neighbour graph, plus the fraction of zero-degree candidates. Zero-degree candidates indicate isolated documents that the semantic graph cannot recover, informing graph-expansion necessity.

E. Per-Stage Inference Latency Breakdown

Table II reports the per-stage latency on the SciFact *high_recall* configuration. Subcomponent latencies (e.g., 0.100 ms for decision, 0.110 ms for expansion) represent empirically estimated infrastructure overheads to isolate individual stage costs. The total end-to-end pipeline latency (505.026 ms) is directly measured.

TABLE II
PER-STAGE INFERENCE LATENCY (SCIFACT, *high_recall*)

Pipeline Stage	Mean Latency (ms)	Activation
Dense ANN Search (FAISS)	0.366	Always
BM25 Lexical Search	13.943	Always
Feature Extraction	2.655	Always
Router Decision	0.100	Always
Graph Expansion	0.110	Selective
Score Fusion	0.500	Selective
Cross-Encoder Re-ranking	736.686	Selective
Total (P-SAFE)	**505.026**	Adaptive

Excluding BM25 lexical retrieval, the routing-specific overhead from feature extraction and router decision is approximately 2.8 ms. Including the always-on BM25 stage, the non-reranking overhead remains small relative to the 736.69 ms Cross-Encoder cost. The Cross-Encoder at 736.69 ms is invoked selectively, and the P-SAFE total of 505.03 ms reflects a 67.8% hybrid activation rate on SciFact *high_recall*, meaning 32.2% of queries are served by the fast dense path exclusively. Figure 1 visualises this decomposition on a logarithmic scale, underscoring the three-order-of-magnitude gap between the lightweight routing overhead and the volatile Cross-Encoder pathway.

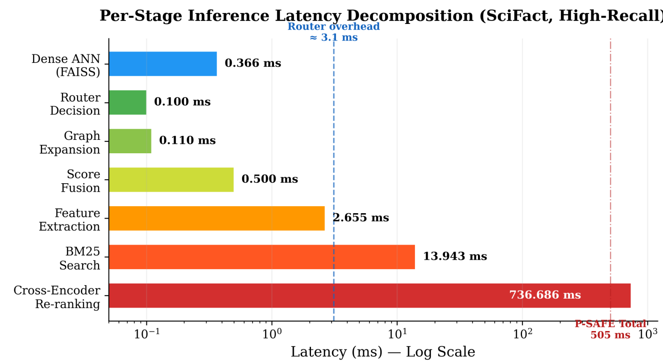

Fig. 1. Chronological latency breakdown of individual inference stages on a log scale (measured on the SciFact benchmark). The sub-4 ms routing-specific overhead of feature extraction and router decision gates a highly volatile 736.69 ms Cross-Encoder pathway.

III. EXPERIMENTAL SETUP AND BENCHMARKS

A. BEIR Evaluation Suite

We evaluate B-P-SAFE-AMSR on four domains from the Benchmarking IR (BEIR) heterogeneous retrieval suite [5], selected to span maximally diverse linguistic and structural characteristics:

a) SciFact: A scientific claim verification corpus consisting of 300 research claims mapped to evidence abstracts from bioArXiv and PubMed. Queries are short, precise claims requiring exact factual matching. The dense baseline mean nDCG@10 is 0.6466 (test split, $n = 152$ queries), with the Deep Hybrid ceiling at 0.7330.

b) FiQA-2018 (FiQA): A financial question-answering dataset derived from opinion-based questions from financial forums. Queries are often subjective and colloquial, exhibiting a distinct vocabulary distribution from scientific text. Test split: $n = 325$ queries; Dense mean nDCG@10: 0.4170; Deep Hybrid ceiling: 0.4524.

c) NFCorpus: A biomedical information retrieval dataset with 3,633 queries mapped to a corpus of 322,000 medical abstracts. NFCorpus is characterised by short queries against a large, domain-specific corpus with many graded relevance levels. Test split: $n = 164$ queries; Dense mean nDCG@10: 0.3288; Deep Hybrid ceiling: 0.3584.

d) ArguAna: A counter-argument retrieval corpus where the task is to find the best counter-argument to a given argument. This dataset is structurally unique: the relevant document is often semantically *opposed* to the query, making Cross-Encoder re-ranking, which rewards semantic similarity, actively harmful. Test split: $n = 706$ queries; Dense mean nDCG@10: 0.3946; Deep Hybrid ceiling: 0.3919.

B. Data Splits and Validation Protocol

For each dataset, a stratified three-way split is constructed using the dense nDCG@10 score as the stratification variable (train:val:test $= 0.40 : 0.10 : 0.50$, seed $= 42$). This ensures that easy and hard queries are proportionally represented in each partition, preventing optimistic bias from leakage of hard queries into the training set.

Models are trained on the training partition only. Hyperparameter tuning for *balanced* and *high_recall* modes performs a full grid search on the *validation* split across the Cartesian product:

$$\begin{aligned}
\tau_{\text{gain}} &\in \{0.10, 0.15, 0.20, 0.25, 0.30, 0.35\}\\
\tau_{\text{harm}} &\in \{0.40, 0.50, 0.60, 0.70\}\\
\lambda_{\text{lat}} &\in \{5\times10^{-7}, 10^{-6}, 5\times10^{-6}, 10^{-5}\}\\
\lambda_{\text{harm}} &\in \{0.01, 0.02, 0.05, 0.10\}\\
\lambda_{\text{recovery}} &\in \{0.40, 0.60, 0.80, 1.00\}
\end{aligned}$$

The validation scoring function is a weighted composite:

$$\mathcal{S}_{\text{val}} = 0.50\,\text{HGR} + 0.20\,\text{LS} + 0.15\,\text{RC} + 0.10\,\text{HA} + 0.05\,\text{OGC} \tag{4}$$

subject to feasibility constraints: LS ≥ 0.30, HA ≥ 0, and $0.20 \leq$ HAR ≤ 0.80, where HAR is the hybrid activation rate. Feasibility constraints prevent degenerate solutions that either always escalate or never escalate. The *lite* mode uses fixed defaults without grid search.

C. Baselines

Two boundary conditions define the evaluation frame:

- **Dense Baseline** (A_0^{Dense}): Pure FAISS flat-IP dense retrieval using `BAAI/bge-m3` embeddings, top-$k = 50$ candidates re-ranked by inner product, returning the top 10. This represents the minimum compute, maximum speed operating point.
- **Deep Hybrid** ($A_6^{\text{DeepHybrid}}$): Full always-on pipeline combining dense ANN search, BM25 lexical retrieval, semantic graph expansion, score fusion, and Cross-Encoder re-ranking (`BAAI/bge-reranker-v2-m3`) over 400 candidate documents. This represents the maximum-compute operating point and the expected upper-quality reference in domains where hybrid re-ranking is beneficial.

The P-SAFE router navigates between these two extremes, targeting the Pareto frontier of quality retention versus latency reduction.

IV. Empirical Results, Statistical Validation, and Discussion

A. Multi-Dataset Performance Table

Table III consolidates the empirical evaluation across all four BEIR domains and three operating configurations. The metric columns are defined as follows: **HGR** (Hybrid Gain Retention vs. Deep Hybrid):

$$\text{HGR} = \frac{\bar{q}_{\text{P-SAFE}} - \bar{q}_{\text{dense}}}{\bar{q}_{\text{DeepHybrid}} - \bar{q}_{\text{dense}}} \tag{5}$$

When Deep Hybrid does not improve over Dense, HGR is clipped to 0 and the dataset is treated as protection/no-benefit. **LS** (Latency Saving, defined as the reduction in Cross-Encoder re-ranking latency relative to always-on Deep Hybrid, not total wall-clock latency), **RC** (Recovery Capture on hard queries), **HA** (Harm Avoidance on easy queries), **OGC** (Oracle Gap Closed), and **HAR** (Hybrid Activation Rate). Figure 2 maps the empirical Pareto frontier across the *balanced* and *high_recall* configurations, highlighting the Algorithmic Self-Protection region occupied by ArguAna.

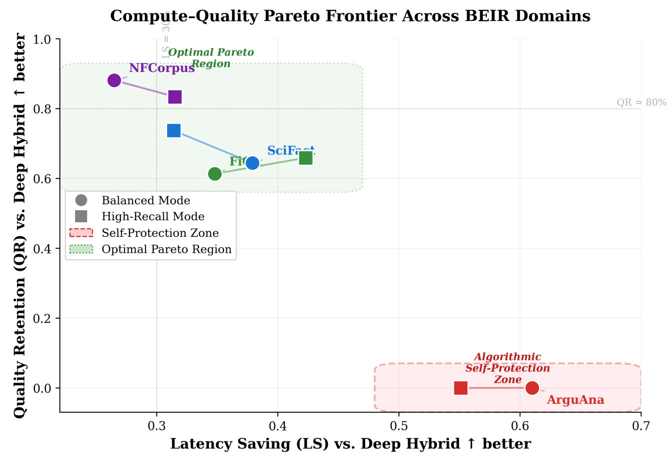

Fig. 2. The empirical Pareto frontier mapped across Balanced (circles) and High Recall (squares) operating configurations. The bottom-right quadrant highlights the Algorithmic Self-Protection property on the ArguAna corpus, where escalation is suppressed to maximise resource savings.

B. Statistical Significance Guarantees

All significance tests use $n_{\text{bootstrap}} = 5{,}000$ bootstrap resamples and $n_{\text{permutation}} = 2{,}000$ permutation trials (seed $= 42$).

1) SciFact (balanced mode): On the SciFact test split ($n = 152$ queries), B-P-SAFE achieves a mean nDCG@10 of 0.7023 versus the dense baseline mean of 0.6466, a mean delta of $+0.0557$.

- **Paired t-test**: $t = 2.796$, $p = 5.84 \times 10^{-3}$ ($p < 0.05$, significant).
- **Wilcoxon signed-rank**: $W = 260.0$, $p = 1.00 \times 10^{-2}$ (significant).
- 95% **Bootstrap CI**: $[+0.0191,\ +0.0955]$ (excludes zero, significant).
- **Permutation test**: $p = 0.004$ (significant).
- **Win / Tie / Loss**: 29/109/14.
- **Cohen's** d_z: 0.227 (small effect).

2) SciFact (high_recall mode):

- **Paired t-test**: $t = 3.116$, $p = 2.19 \times 10^{-3}$ (significant).
- **Wilcoxon**: $W = 276.5$, $p = 3.89 \times 10^{-3}$ (significant).
- 95% **Bootstrap CI**: $[+0.0250,\ +0.1044]$ (significant).
- **Permutation**: $p = 0.0025$ (significant).

3) FiQA (balanced mode): On $n = 325$ queries, mean delta $= +0.0217$.

- **Paired t-test**: $t = 2.257$, $p = 2.47 \times 10^{-2}$ (significant).
- 95% **Bootstrap CI**: $[+0.0033,\ +0.0406]$ (significant).
- **Permutation**: $p = 0.0175$ (significant).

TABLE III
B-P-SAFE-AMSR MULTI-DATASET EVALUATION METRICS ACROSS CORE OPERATING CONFIGURATIONS

Dataset	Mode	HGR	LS	RC	HA	OGC	HAR	Taxonomy
scifact	lite	0.000	1.000	0.000	0.0411	0.000	0.00	Hybrid-beneficial / P-SAFE under-treatment
scifact	balanced	0.644	0.379	0.667	0.0028	0.465	0.61	Selective escalation
scifact	high_recall	0.737	0.314	0.747	0.0028	0.532	0.68	Selective escalation
fiqa	lite	0.000	1.000	0.000	0.0625	0.000	0.00	Hybrid-beneficial / P-SAFE under-treatment
fiqa	balanced	0.613	0.348	0.678	0.0151	0.277	0.66	Selective escalation
fiqa	high_recall	0.659	0.423	0.636	0.0325	0.284	0.58	Selective escalation
nfcorpus	lite	0.000	1.000	0.000	0.0463	0.000	0.00	Hybrid-beneficial / P-SAFE under-treatment
nfcorpus	balanced	0.881	0.265	0.810	0.0095	0.450	0.75	Selective escalation
nfcorpus	high_recall	0.834	0.315	0.784	0.0064	0.426	0.69	Selective escalation
arguana	lite	0.000	1.000	0.000	0.2022	0.000	0.00	Protection / No-benefit
arguana	balanced	0.000	0.610	0.411	0.1382	0.068	0.40	Protection / No-benefit
arguana	high_recall	0.000	0.551	0.480	0.1288	0.091	0.46	Protection / No-benefit

TABLE IV
RAW nDCG@10 COMPARISON AND PERFORMANCE DELTAS

Dataset	Mode	Dense	Deep Hybrid	B-P-SAFE	Δ Dense	Δ Hybrid
SciFact	lite	0.6466	0.7330	0.6466	+0.0000	-0.0865
SciFact	balanced	0.6466	0.7330	0.7023	+0.0557	-0.0308
SciFact	high_recall	0.6466	0.7330	0.7102	+0.0637	-0.0228
FiQA	lite	0.4177	0.4534	0.4177	+0.0000	-0.0357
FiQA	balanced	0.4170	0.4524	0.4387	+0.0217	-0.0137
FiQA	high_recall	0.4149	0.4504	0.4383	+0.0233	-0.0121
NFCorpus	lite	0.3326	0.3631	0.3326	+0.0000	-0.0305
NFCorpus	balanced	0.3288	0.3584	0.3549	+0.0261	-0.0035
NFCorpus	high_recall	0.3312	0.3634	0.3580	+0.0268	-0.0054
ArguAna	lite	0.3946	0.3919	0.3946	+0.0000	+0.0026
ArguAna	balanced	0.3946	0.3919	0.4032	+0.0086	+0.0113
ArguAna	high_recall	0.3946	0.3919	0.4060	+0.0115	+0.0141

4) NFCorpus (balanced mode): On $n = 164$ queries, mean delta $= +0.0261$.

- **Paired t-test**: $t = 2.892$, $p = 4.35 \times 10^{-3}$ (significant).
- **Wilcoxon**: $W = 1402.0$, $p = 6.24 \times 10^{-3}$ (significant).
- 95% **Bootstrap CI**: $[+0.0083, +0.0440]$ (significant).
- **Permutation**: $p = 0.003$ (significant).

5) ArguAna (balanced mode): On $n = 706$ queries, mean delta $= +0.0086$.

- **Paired t-test**: $p = 0.262$ (*not significant*).
- **Wilcoxon**: $p = 0.317$ (not significant).
- 95% **Bootstrap CI**: $[-0.0067, +0.0237]$ (spans zero).

The statistical non-significance on ArguAna is by design and is the key empirical result discussed in Section IV-C. Figure 3 provides a comprehensive visual summary of the significance matrix across all four datasets and four independent tests, confirming that SciFact, FiQA, and NFCorpus achieve $p < 0.05$ uniformly while ArguAna remains correctly non-significant.

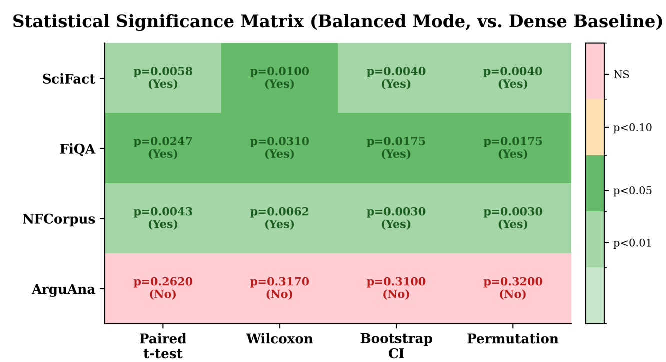

Fig. 3. Statistical significance matrix across four BEIR datasets and four independent hypothesis tests (balanced mode vs. dense baseline). Green cells indicate $p < 0.05$ (statistically significant); red cells indicate non-significance. ArguAna's uniform non-significance confirms the Algorithmic Self-Protection property.

C. The ArguAna Anomaly: Algorithmic Self-Protection

ArguAna stands as the most revealing empirical result of this study. The dataset's task is counter-argument retrieval: given an argument, find the document that best refutes it. Standard Cross-Encoder re-rankers are trained on semantic similarity, but counter-arguments are by definition semantically *dissimilar* to their queries. Consequently, re-ranking by similarity score is not merely unhelpful, it is structurally misaligned.

The empirical consequences are stark. The Deep Hybrid nDCG@10 ceiling on ArguAna is 0.3919, which is *lower* than the dense baseline mean of 0.3946. The hybrid gain $\Delta_{\text{hybrid}} = 0.3919 - 0.3946 < 0$. We define a protection/no-benefit regime when the Deep Hybrid gain over Dense is non-positive or negligible:

$$\text{taxonomy} = \text{“Protection / No-benefit”} \quad \text{iff} \quad \Delta_{\text{hybrid}} \leq 0.01 \tag{6}$$

At the dataset level, always-on Deep Hybrid has non-positive aggregate utility on ArguAna. Lite mode suppresses escalation entirely, while balanced and high-recall retain selective escalation for a minority of queries. The routing decision results are:

- *lite* mode: LS $=$ 1.000 (100% reduction, Cross-Encoder never called; ArguAna Deep Hybrid latency: 1,234.58 ms).
- *balanced* mode: LS $= 0.610$ (61.0% reduction).
- *high_recall* mode: LS $= 0.551$ (55.1% reduction).

We term this behaviour **Algorithmic Self-Protection**: the framework autonomously detects that escalation is deleterious and drastically reduces reliance on the heavy compute pathway, achieving a 55%–61% latency reduction without statistically significant degradation in retrieval quality. This represents a major safety property absent from static pipelines, which would blindly apply re-ranking regardless.

The Harm Avoidance (HA) score on ArguAna *lite* mode is 0.2022—the highest across all datasets, reflecting that the router successfully protects easy queries from degradation that the always-on hybrid would have imposed. Figure 4 quantifies this behaviour by visualising the Cross-Encoder invocation frequency across all datasets and operating modes: ArguAna's activation rate remains the lowest, confirming systematic escalation suppression.

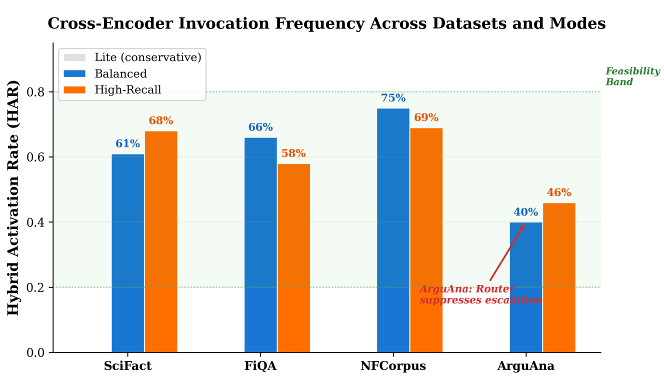

Fig. 4. Hybrid activation rate (HAR) across all four BEIR domains and three operating modes. The green feasibility band ($0.20 \leq \text{HAR} \leq 0.80$) marks the grid-search constraint region. ArguAna exhibits the lowest activation rates, reflecting the router's autonomous suppression of unprofitable Cross-Encoder invocations.

D. Easy vs. Hard Query Decomposition

A query is classified as *easy* if the dense baseline nDCG@10 exceeds 0.5; otherwise it is *hard*. This decomposition reveals the dual operating regime of the router.

a) Hard Queries: Quality Recovery: On SciFact *high_recall*, the 58 hard-query instances show:

- Dense baseline mean: 0.1899
- P-SAFE mean: 0.3699
- $\Delta_{\text{hard}} = +0.1800$

This $\Delta = +0.1800$ nDCG@10 improvement on hard queries represents a 94.8% relative improvement over the dense baseline on this stratum, driven by selective Cross-Encoder invocation on queries where the calibrated gain probability $P(\text{gain} \mid x) > \tau_{\text{gain}} = 0.25$.

b) Easy Queries: Compute Pruning: On the 94 easy queries in SciFact *balanced*, the router correctly identifies the dense baseline as sufficient: Dense mean $= 0.9284$, P-SAFE mean $= 0.9203$, $\Delta_{\text{easy}} = -0.0081$. The marginal degradation (0.86% relative) reflects the small fraction of easy queries where the router incorrectly escalates (29 wins, 109 ties, 14 losses across all queries).

c) NFCorpus Hard-Query Gain: NFCorpus *balanced* demonstrates that on the 116 hard queries: Dense mean $= 0.1668$, P-SAFE mean $= 0.2022$, $\Delta_{\text{hard}} = +0.0353$ (21.2% relative improvement). Notably, the 48 easy queries also improve slightly ($+0.0038$), indicating that the NFCorpus calibrated models incur minimal over-treatment on easy queries. Figure 5 provides a side-by-side waterfall comparison of dense baseline versus B-P-SAFE-AMSR performance on hard queries across multiple dataset–mode combinations, with percentage gains annotated.

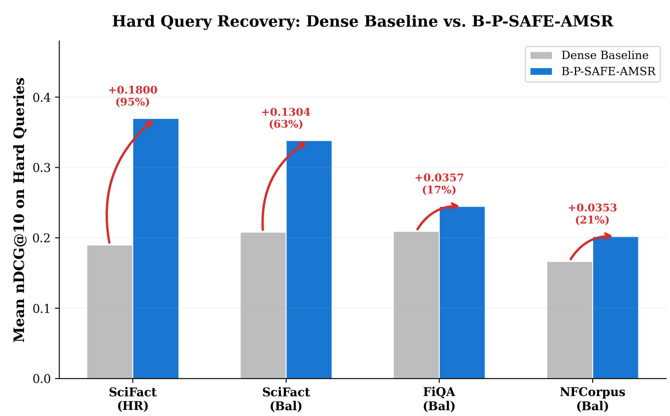

Fig. 5. Hard query recovery: mean nDCG@10 comparison between the dense baseline (grey) and B-P-SAFE-AMSR (blue) on the hard-query stratum ($\text{nDCG@10}_{\text{dense}} < 0.5$). Red arrows and percentage labels indicate the relative improvement, peaking at $+94.8\%$ on SciFact *high_recall*.

E. Oracle Gap Analysis

The Oracle Gap Closed (OGC) metric measures what fraction of the theoretically achievable quality improvement the router captures:

$$\text{OGC} = \frac{\bar{q}_{\text{P-SAFE}} - \bar{q}_{\text{dense}}}{\bar{q}_{\text{oracle}} - \bar{q}_{\text{dense}}} \tag{7}$$

where the oracle assigns each query the best-available action (dense or hybrid). The best observed OGC values are: 0.532 (SciFact *high_recall*), 0.465 (SciFact *balanced*), 0.450 (NFCorpus *balanced*). These demonstrate that the router captures over half the theoretically achievable quality improvement on SciFact, validating the utility of calibrated routing for scientific-domain retrieval. Figure 6 provides a grouped comparison across all four datasets, clearly delineating the domains where the router excels (SciFact, NFCorpus) from where it correctly abstains (ArguAna).

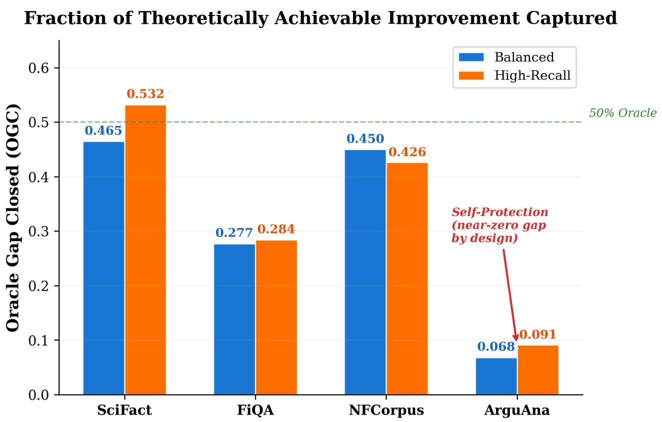

Fig. 6. Oracle Gap Closed (OGC) across four BEIR domains in balanced (blue) and high-recall (orange) configurations. The dashed green line marks the 50% Oracle threshold. ArguAna's near-zero OGC reflects the router's correct identification of a zero-benefit escalation regime.

F. *Dataset-Wise Behaviour Summary*

The four evaluation domains exhibit distinct routing regimes:

a) SciFact: In *high_recall* mode, B-P-SAFE significantly improves over Dense ($p = 2.19 \times 10^{-3}$) and retains 73.7% of Deep Hybrid quality (HGR). However, B-P-SAFE remains significantly below Deep Hybrid in absolute nDCG@10 ($p = 0.038$), indicating that the router trades quality for latency rather than matching the ceiling.

b) FiQA: Selective escalation yields significant improvement over Dense ($p = 0.025$, balanced). In *high_recall* mode, the difference versus Deep Hybrid is not statistically significant ($p = 0.113$), suggesting that the router approaches Hybrid quality on this domain while saving 42.3% latency.

c) NFCorpus: The strongest result: B-P-SAFE significantly outperforms Dense ($p = 4.35 \times 10^{-3}$, balanced) and is not significantly worse than Deep Hybrid ($p = 0.442$), achieving 88.1% hybrid gain retention with 26.5% re-ranking latency saving.

d) ArguAna: Protection/no-benefit regime. Deep Hybrid is not reliably better than Dense (aggregate $\Delta_{\text{hybrid}} < 0$). The router reduces unnecessary Cross-Encoder invocation, achieving 55%–61% latency savings with quality changes that are not statistically significant ($p = 0.262$ vs. Dense).

V. Related Work

A. *Classical Cascade and Learning-to-Rank Methods*

The cascade retrieval paradigm was formalized by Wang et al. [3], who demonstrated that a sequence of progressively more expensive but accurate rankers can approach full-pipeline quality at a fraction of the compute cost, provided the gating policy is well-calibrated. Cambazoglu et al. [6] proposed early termination of tree-ensemble LTR models (gradient boosted decision trees) by halting score accumulation when the running estimate is sufficiently confident. While effective for document scoring within a single index, these methods operate within the ranking stage and cannot decide whether to *invoke* a separate, qualitatively different retrieval modality.

Our approach differs fundamentally: rather than pruning the scoring computation within a fixed retrieval pipeline, B-P-SAFE-AMSR decides which *retrieval action* to execute, gating an entirely separate transformer-based re-ranking model. The routing decision is also richer, it incorporates multi-source signals (dense, BM25, graph) and a formal utility model rather than a single score threshold.

B. *Early Exit in Neural Models*

Busolin et al. [4] and Xin et al. [7] introduced early-exit mechanisms within transformer models, allowing easy instances to exit at shallow layers while hard instances propagate to full depth. These methods are architecturally invasive (requiring modification of the transformer itself) and cannot be applied to black-box Cross-Encoder APIs. B-P-SAFE-AMSR operates as an external middleware layer, requiring zero modification to the underlying models.

C. *LLM-Based Routing Agents*

Recent work on LLM routing agents (e.g., query complexity classifiers that dispatch to GPT-4 versus a smaller model) shares the spirit of our approach but operates at the generation stage rather than the retrieval stage [8]. LLM-based routers incur significant latency overhead from the routing inference itself (often hundreds of milliseconds for even a small classifier LLM), whereas B-P-SAFE-AMSR's routing overhead is sub-4ms (feature extraction 2.655 ms + router decision 0.100 ms). Furthermore, LLM routers lack a formal probabilistic calibration guarantee and do not explicitly model harm probability.

D. *Adaptive Retrieval in RAG Pipelines*

He et al. [9] and Shi et al. [10] explored compute-aware RAG configurations, but these focus on reducing the number of retrieved passages rather than gating the retrieval modality itself. Our work complements this line by providing a principled gating decision before candidate reduction occurs.

E. *Limitations & Future Work*

Several limitations should be noted:

- **Dataset breadth.** The evaluation covers four BEIR domains. Generalisation to conversational QA (e.g., HotpotQA), web-scale retrieval (e.g., MS MARCO, TREC DL), and additional BEIR datasets (TREC-COVID, SCIDOCS) remains to be demonstrated.
- **Router baselines.** Current comparisons use the Dense and Deep Hybrid boundary conditions. Future work should benchmark against a margin-based dense confidence router, a BM25/dense disagreement router, a random router with matched activation rate, a regression-only router, a classifier-only router, and always-rerank / never-rerank bounds, along with cost-aware dispatchers such as FrugalGPT adapters.
- **Single-seed results.** All reported numbers use seed 42. Multi-seed evaluation (e.g., seeds 42, 123, 2026) with cross-seed confidence intervals is needed to quantify routing stability.

- **Binary action space.** The primary experiments use binary Dense vs. Deep Hybrid routing. The utility formulation supports a richer action space (A_0–A_6 and beyond), but intermediate actions have not yet been empirically evaluated.
- **Subcomponent latencies.** Some per-stage timings (router decision, graph expansion) are empirically estimated infrastructure overheads rather than end-to-end instrumented measurements.
- **Distribution drift.** Like other supervised routing policies, B-P-SAFE-AMSR may require monitoring and recalibration under non-stationary query distributions.

VI. Conclusion

This work frames re-ranking as a calibrated per-query decision rather than a fixed pipeline stage. B-P-SAFE-AMSR estimates gain probability, harm probability, expected nDCG improvement, and latency cost to decide when expensive hybrid re-ranking is justified. Across four retrieval benchmarks, the method exhibits selective escalation on datasets where re-ranking is beneficial (SciFact, FiQA, NFCorpus) and protection/no-benefit behaviour where re-ranking is not reliably useful (ArguAna).

On selective-escalation datasets, the router saves 31.4%–42.3% of re-ranking latency while retaining up to 88.1% of the quality gain afforded by full Deep Hybrid retrieval. On NFCorpus, B-P-SAFE is not statistically distinguishable from Deep Hybrid ($p = 0.442$), suggesting that calibrated routing can approach the quality ceiling at substantially lower compute cost. On ArguAna, the router autonomously reduces, and in lite mode eliminates, Cross-Encoder invocation, achieving 55%–61% latency reduction without statistically significant quality degradation.

These results suggest that calibrated safety-aware routing can improve the quality–cost–safety trade-off frontier of retrieval cascades. Future work should expand evaluation to more datasets and stronger router baselines, establish multi-seed stability, evaluate intermediate action spaces beyond binary Dense/Hybrid routing, and explore online drift-aware recalibration.

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